Phase 1: Problem Statements & Scope Definition

Smart Healthcare Risk & Chronic Disease Surveillance Capstone Project

1. Primary Problem Statement

Clinical practitioners and healthcare administrators lack unified, real-time visual analytics systems that translate complex biometric data (age, blood pressure, fasting glucose, cholesterol, BMI, lifestyle factors) into calibrated, reproducible chronic disease risk scores.

2. Specific Research & Clinical Questions

- At what age inflection point do cardiovascular disease, diabetes, and stroke rates surge in the patient population?
- What is the true prevalence of triad multi-morbidity in the cohort, and how does it correlate with biomarker elevation?
- How can data quality anomalies (such as constant values in raw risk scores) be systematically identified and rectified?

3. Project Boundary & Scope

- Target Population: 5,000 adult patient records (ages 18 to 79).
- Target Diseases: Heart Disease, Diabetes Mellitus, and Stroke.
- Modifiable Factors: Smoking status, alcohol intake, and physical exercise frequency.